

Customer Segmentation Project

Objective: Segment customers based on behavior and demographics using Python and K-Means clustering.

Dataset: 500 customers **Best K:** 2 **Silhouette Score:** 0.223

1. Business Problem

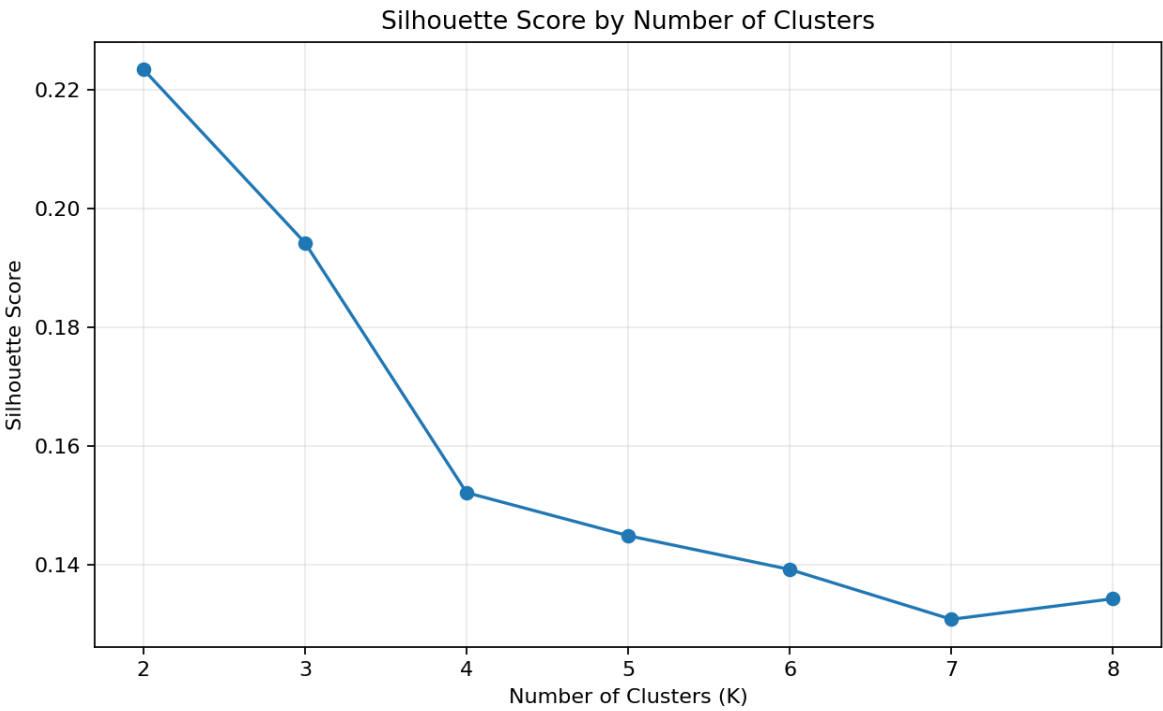
Customer segmentation groups customers with similar purchasing and engagement behavior, enabling targeted marketing, retention and personalization.

2. Data Preparation

Missing numeric values were replaced with median values. Demographic and behavioral variables were selected and standardized using StandardScaler.

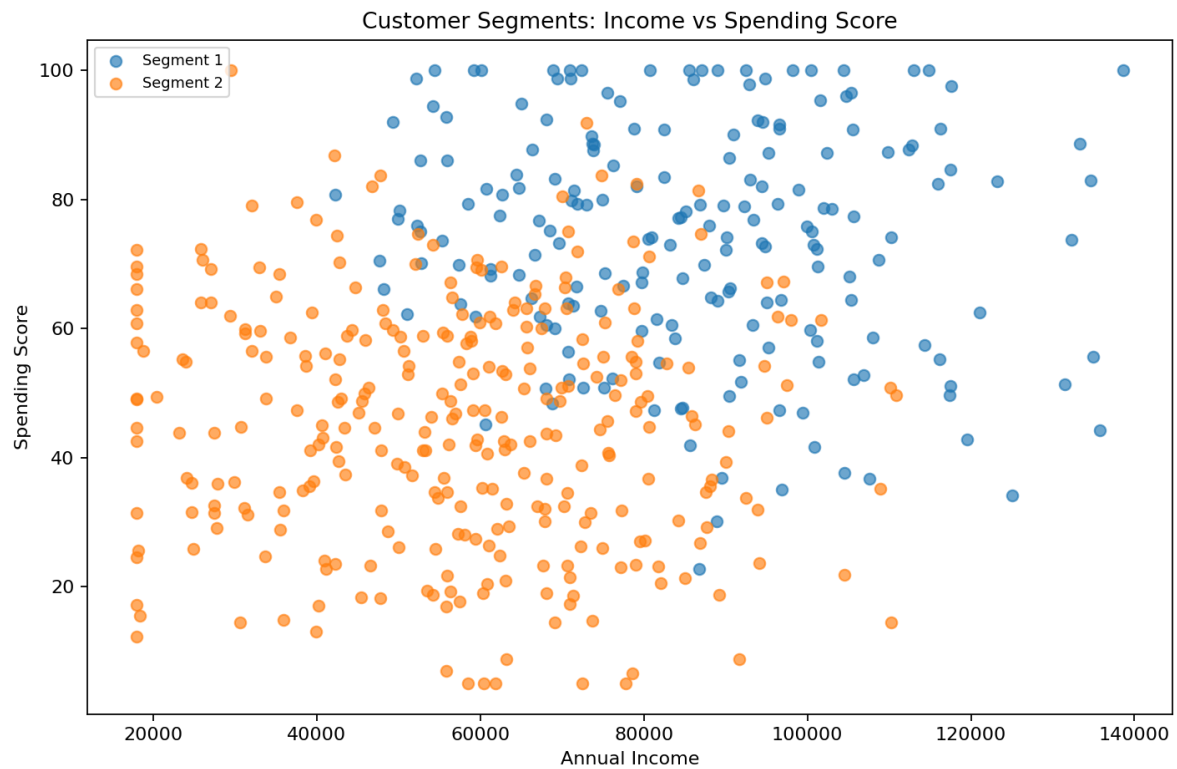
3. Clustering Method

K-Means was tested for K=2 through K=8. The highest silhouette score was 0.223 at K=2, which was selected for the final model.



4. Segment Profile

Segment	Customers	Avg Income	Avg Spend	Avg Purchases	Avg Recency
Segment 2	305	\$57,073	45.2	7.5	51.6
Segment 1	195	\$85,924	73.9	12.7	25.3



5. Business Recommendations

High-Value Customers: retention, loyalty rewards and premium offers.
Loyal Regulars: personalized recommendations and repeat-purchase incentives.
Budget-Conscious Customers: value bundles and selective discounts.
At-Risk Customers: win-back and re-engagement campaigns.

6. Conclusion

The project demonstrates a complete customer analytics workflow from data preprocessing through clustering, visualization and actionable insights.